The Bloch Sphere

1 What Is It?

A qubit state is written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

The two complex numbers α and β look like four real parameters, but the normalisation constraint and the fact that global phase is unobservable each remove one, leaving exactly **two free parameters**. Two parameters describe a point on a 2D surface in 3D space — and that surface is a unit sphere. Felix Bloch formalised this in 1946. The Bloch sphere is simply a way to draw every possible single-qubit state.

2 The Angles θ and ϕ

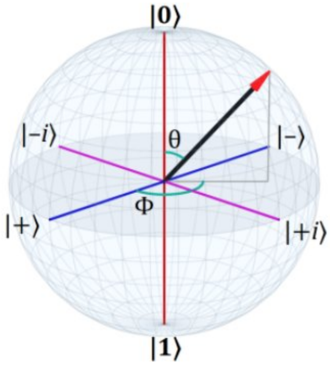

Figure 1: The Bloch sphere. θ is the polar angle from $|0\rangle$; ϕ is the azimuthal angle around the z-axis.

Any state maps to a point on the sphere via

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \qquad \theta \in [0,\pi], \quad \phi \in [0,2\pi).$$

Angle	Range	Controls
θ	$0 \to \pi$	North–south latitude: how much $\|0\rangle$ vs $\|1\rangle$
ϕ	$0 \to 2\pi$	Longitude: relative phase of $\|1\rangle$

The six states you will see everywhere are the poles and equator crossings:

State	θ	ϕ	Location
$\|0\rangle$	0	—	North pole
$\|1\rangle$	π	—	South pole
$\|+\rangle$	$\pi/2$	0	Front of equator
$\|-\rangle$	$\pi/2$	π	Back of equator
$\|+i\rangle$	$\pi/2$	$\pi/2$	Right of equator
$\|-i\rangle$	$\pi/2$	$3\pi/2$	Left of equator

3 Gates and Measurement

Every single-qubit gate is a **rotation** of the Bloch vector. The X gate flips the vector over the x-axis ($|0\rangle \to |1\rangle$). The Z gate spins it around the z-axis (a phase flip). The Hadamard H swaps the poles with the equator ($|0\rangle \to |+\rangle$).

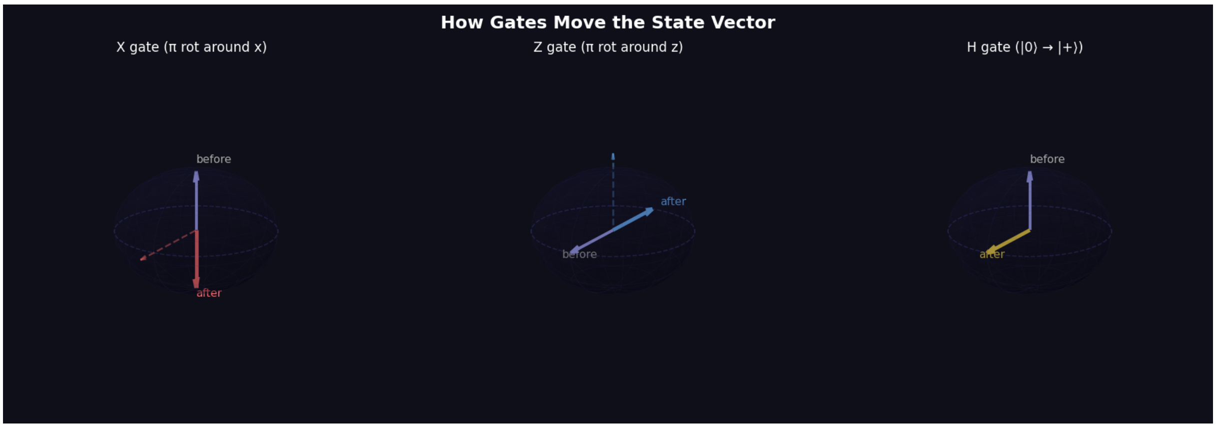

Figure 2: X, Z and H gates as rotations of the Bloch vector.

Measuring in the computational basis collapses the vector to one of the two poles. The probability of each outcome depends only on θ:

$$P(|0\rangle) = \cos^2\left(\frac{\theta}{2}\right), \qquad P(|1\rangle) = \sin^2\left(\frac{\theta}{2}\right).$$

The collapse is irreversible. A state on the equator ($\theta = \pi/2$) gives 50/50.

4 Common Misconceptions

The half-angle is not a typo. The formula uses $\theta/2$, so a full 2π rotation of the Bloch vector corresponds to a 4π rotation of the quantum state. This is spinor behaviour, and it is a genuine property of quantum mechanics, not a convention.

Global phase is unobservable. $e^{i\alpha}|\psi\rangle$ and $|\psi\rangle$ sit at exactly the same point on the sphere. No experiment can distinguish them. Only the relative phase ϕ between $|0\rangle$ and $|1\rangle$ has physical meaning.

The sphere only works for one qubit. Entangled two-qubit states cannot be drawn on a Bloch sphere. Each qubit can still be assigned a reduced state, but it will generally be a mixed state represented by a point *inside* the sphere, not on its surface.

Interactive simulator: https://quantum-orbit-studio.vercel.app/